

From Long to Short: How Interest Rates Shape Life Insurance Markets

Ziang Li

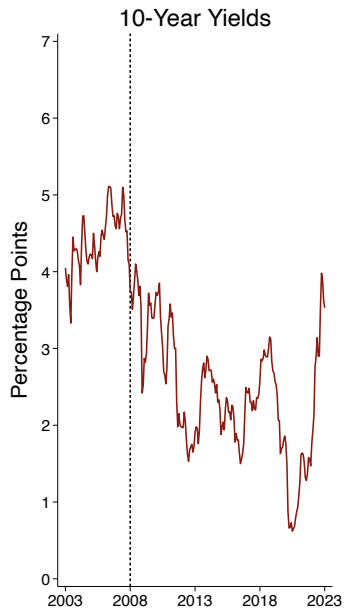
Imperial College

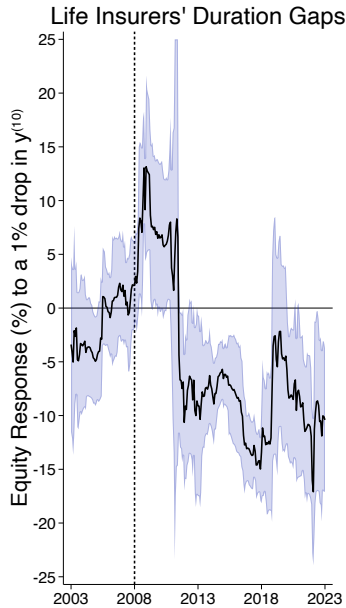
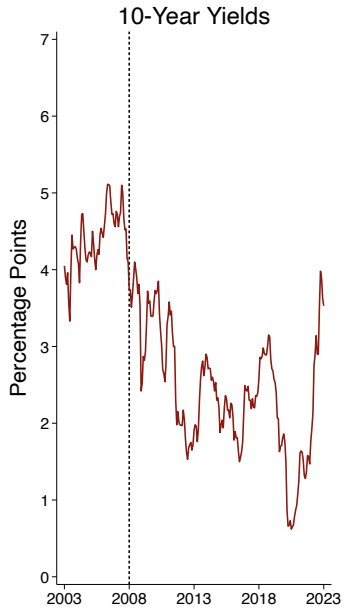
Derek Wenning

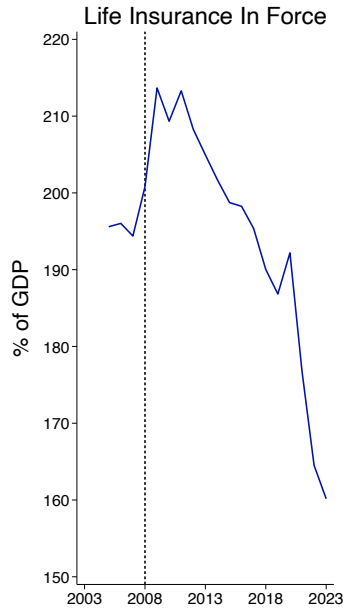
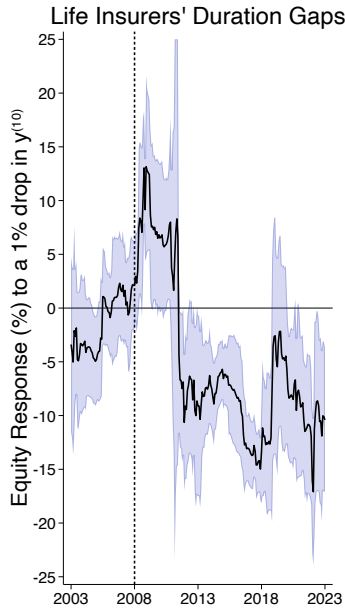
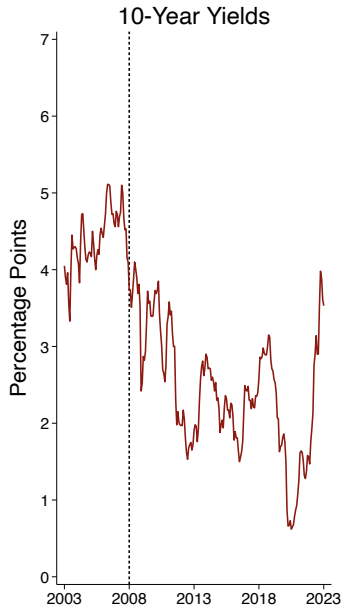
Indiana Kelley

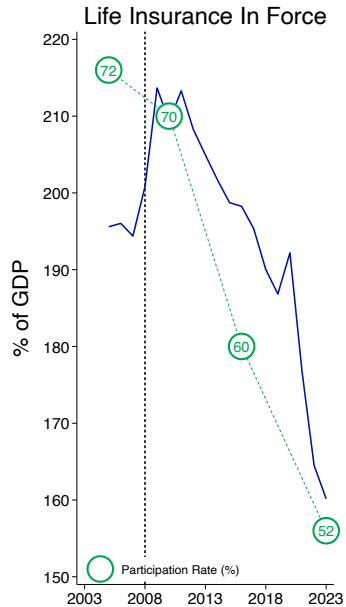
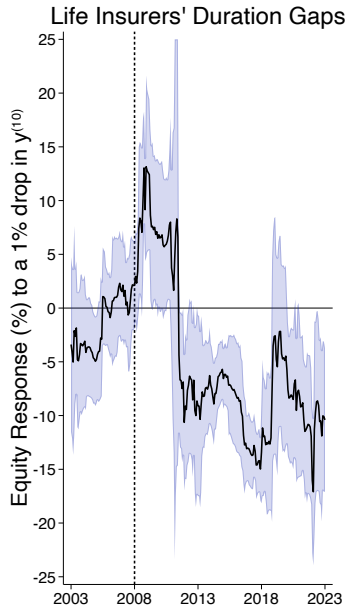
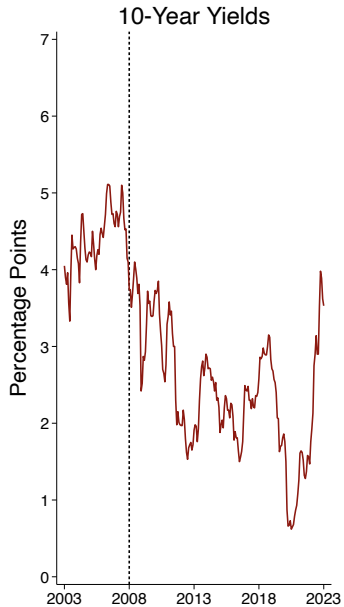
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General Question:

How do the risk management practices of financial institutions affect product characteristics and financial participation?

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This Paper:

How does **interest rate risk** (management) of **life insurers** affect the composition and availability of **life insurance products**?

Main Results: Theoretical + Empirical

- Build a tractable model of insurance product markets + risk management
 - Key mechanism: life insurer risk aversion $\implies$ **prices reflect duration mismatch**
 - Negative duration gap $\implies$ *penalize* long-term products, *subsidize* short-term products

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 - Contrast **VA issuers (exposed)** with **non-VA issuers (non-exposed)**
 - Focus on the post-GFC period when duration mismatch is highest

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 - Contrast **VA issuers (exposed)** with **non-VA issuers (non-exposed)**
 - Focus on the post-GFC period when duration mismatch is highest
- Document several novel findings consistent with our theory
 1. **Duration gaps** turned negative post-2010, especially for exposed insurers
 2. **Prices** increase more for long-term products and exposed insurers
 3. **Quantities (issuance)** shift to favor short-term products $\rightarrow$ liability rebalancing
 4. **Aggregate life insurance (issuance + in force)** shrinks relative to GDP

Theory

Environment: Insurers j , products $i \in \{s, \ell\}$, time $t \in \mathbb{N}$.

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Balance sheet

Capital: $K_{jt} = A_{jt} - L_{jt}$

Returns: $R_{jt}^A = \bar{R}_{jt}^A - D_{jt}^A \Delta R_t$

$$R_{jt}^L = \bar{R}_{jt}^L - D_{jt}^L \Delta R_t$$

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New issuance

Flows: Premiums — $P_{ijt} Q_{ijt}$

Reserves — $V_{it} Q_{ijt}$

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Objective

$$\max_{\{P_{ijt}\}} \underbrace{\sum_i (P_{ijt} - V_{ijt}) Q_{ijt} (P_{ijt})}_{\text{new capital from issuance}} + \underbrace{E_t \left[\Lambda_{jt} \left(R_{jt}^K - E_t [R_{jt}^K] \right) \right]}_{\text{risk management}}$$

Risk Management Motives Affects Optimal Price Setting

$$\log \frac{P_{ijt}}{V_{it}} \approx \underbrace{\log \mu_{it}}_{\text{product mkt markup}} + \underbrace{\mathcal{M}_{ijt}}_{\text{risk mgmt markup}}$$

Risk Management Motives Affects Optimal Price Setting

$$\log \frac{P_{ijt}}{V_{it}} \approx \underbrace{\log \mu_{it}}_{\text{product mkt markup}} + \underbrace{\mathcal{M}_{ijt}}_{\text{risk mgmt markup}} \left\{ \bar{\lambda}'_{jt} \right. \quad \text{(Insurer risk aversion)}$$

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(Insurer risk aversion)
 (Interest rate uncertainty)
(Insurer's duration gap)

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Intuition:

- If $D_{jt}^K = 0$, no need for risk management

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Liability Rebalancing: $D_{jt}^K \downarrow \implies Q_{sjt} \uparrow \text{ and } Q_{\ell jt} \downarrow$

Insurance Supply Shifts more for Insurers with more Convexity

- Consider two otherwise identical insurers, j and j' , where j' has more **convex capital**:

$$\frac{\partial D_{j't+1}^K}{\partial R_{t+1}} < \frac{\partial D_{jt+1}^K}{\partial R_{t+1}} < 0$$

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- Suppose initially that $D_{jt}^K = D_{j't}^K \leq 0$. Then:

$$R_{t+1} \downarrow$$

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$$R_{t+1} \downarrow \implies D_{jt+1}^K \downarrow, D_{j't+1}^K \downarrow\downarrow$$

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$$\frac{\partial D_{j't+1}^K}{\partial R_{t+1}} < \frac{\partial D_{jt+1}^K}{\partial R_{t+1}} < 0$$

- Suppose initially that $D_{jt}^K = D_{j't}^K \leq 0$. Then:

$$R_{t+1} \downarrow \implies D_{jt+1}^K \downarrow, D_{j't+1}^K \downarrow\downarrow \implies \frac{Q_{sjt+1}}{Q_{sjt+1} + Q_{\ell jt+1}} \uparrow, \frac{Q_{sj't+1}}{Q_{sj't+1} + Q_{\ell j't+1}} \uparrow\uparrow$$

Implication: Liability rebalancing is stronger for **more convex insurers**
(e.g., variable annuity issuers)

Broad Changes in Net Duration Can Expand or Contract Product Markets

- With logit demand, (new customer) participation rate for product i is

$$\mathcal{P}_{it} = \frac{\sum_j \alpha_{jt} \mu_{jt}^{1-\varepsilon_{it}} \mathcal{M}_{ijt}^{1-\varepsilon_{it}}}{\alpha_{it}^0 + \sum_j \alpha_{jt} \mu_{jt}^{1-\varepsilon_{it}} \mathcal{M}_{ijt}^{1-\varepsilon_{it}}}$$

Impact on Market-wide Participation:

If $D_{jt}^K \downarrow$ (weakly) for all insurers, participation rate $\mathcal{P}_{st}^0 \uparrow$ and $\mathcal{P}_{\ell t}^0 \downarrow$

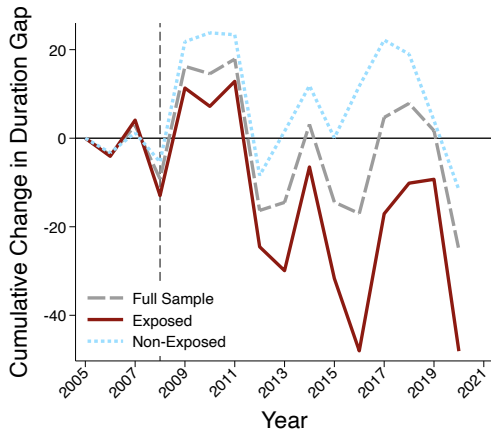
(*Total* participation depends on relative market sizes, distribution costs, etc.)

Data and Empirical Facts

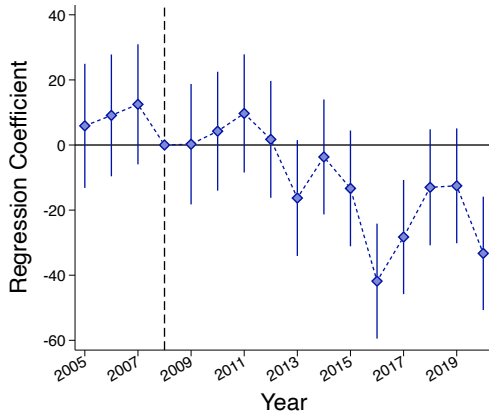
Duration Gaps were Exacerbated Only for Exposed Insurers

- **Previously:** negative duration gaps suggested by stock returns of public insurers
- **Exposed Insurers:** top 10% of (relative) variable annuity liabilities pre-GFC
 - Relatively large (assets \$95B vs. \$8.3B) and levered (19.62 vs. 6.56)
 - Similar market shares across products (43% vs. 54%)
- **Duration Gap** = $D_{jt}^A + \text{LevRatio}_{jt}(D_{jt}^A - D_{jt}^L)$
 - D_{jt}^A : asset duration, approximated by corporate bond duration
 - D_{jt}^L : liability duration, taken from Huber (2022)
 - LevRatio_{jt} : Liabilities / *Surplus* Capital
- **Regression Analysis:**
$$D_{jt}^K = \sum_{\tau=2005}^{2020} \beta_{\tau} \mathbf{1}\{t = \tau\} \times \text{Exposed}_j + \delta_j + \delta_t + \varepsilon_{jt}$$

Duration Gaps were Exacerbated Only for Exposed Insurers



(a) Duration Gap Changes, Raw Data



(b) Gap Differences, Regression Output

How Does Duration Mismatch Affect Product Pricing?

$$\log \text{Price}_{ijt} = \beta \times y_t^{(10)} \times \text{Exposed}_j \times \text{Long}_i + \delta_{jt} + \delta_{it} + \delta_{ij} + \varepsilon_{ijt}$$

- Prices are 10, 15, or 20-year term life quotes at a monthly frequency
- $\beta < 0$: exposed insurers mark up long products relatively more when rates are low
- Fixed effects control for several alternative stories
 - δ_{jt} — absorbs the impacts of insurer characteristics (e.g., size, leverage) on product supply
 - δ_{it} — absorbs the time-varying demand for a specific product category i
 - δ_{ij} — absorbs heterogeneous insurer productivity/characteristics across products

Exposed Insurers Move Long-Short Product Spreads More

$$\log \text{Price}_{ijt} = \beta \times y_t^{(10)} \times \text{Exposed}_j \times \text{Long}_i + \delta_{jt} + \delta_{it} + \delta_{ij} + \varepsilon_{ijt}$$

<i>(Long, Short) Category:</i>	(15,10)	(20,15)	(20,10)
$y_t^{(10)} \times \text{Exposed}_j \times \text{Long}_i$	-0.006*** (0.002)	-0.018*** (0.002)	-0.023*** (0.003)

Insurer $\times$ Month FE	✓	✓	✓
Insurer $\times$ Product FE	✓	✓	✓
Month $\times$ Product FE	✓	✓	✓

Observations	8956	8956	8956
Within- R^2	0.001	0.023	0.020

Exposed Insurers Move Long-Short Product Spreads More

$$\log \text{Price}_{ijt} = \beta \times \text{MPU}_t \times \text{Exposed}_j \times \text{Long}_i + \delta_{jt} + \delta_{it} + \delta_{ij} + \varepsilon_{ijt}$$

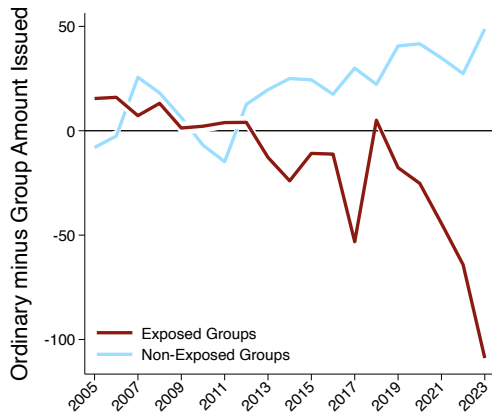
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$\text{MPU}_t \times \text{Exposed}_j \times \text{Long}_i$				0.007*** (0.002)	0.013*** (0.002)	0.020*** (0.003)
Insurer $\times$ Month FE	✓	✓	✓	✓	✓	✓
Insurer $\times$ Product FE	✓	✓	✓	✓	✓	✓
Month $\times$ Product FE	✓	✓	✓	✓	✓	✓
Observations	8956	8956	8956	8956	8956	8956
Within- R^2	0.001	0.023	0.020	0.002	0.014	0.017

How Do Pricing Distortions Affect Quantities?

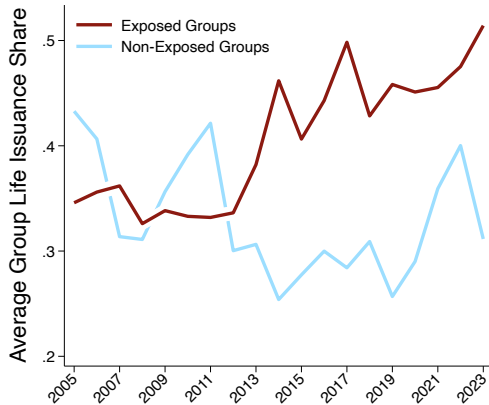
- **Ordinary Life:** long term (term or whole life), accessible through agents
 - Interest sensitive due to whole life guarantees, surrender/lapsation risk
- **Group Life:** yearly renewable, accessible through employers
 - No dynamic component → little to no duration

Theory: Exposed insurer duration gaps $\uparrow \rightarrow Q_{jt}^{\text{group}} \uparrow$ while $Q_{jt}^{\text{ordinary}} \downarrow$

Exposed Insurers Transition to Short Term Group Policies



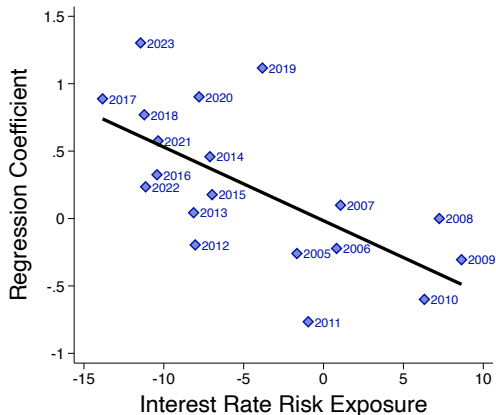
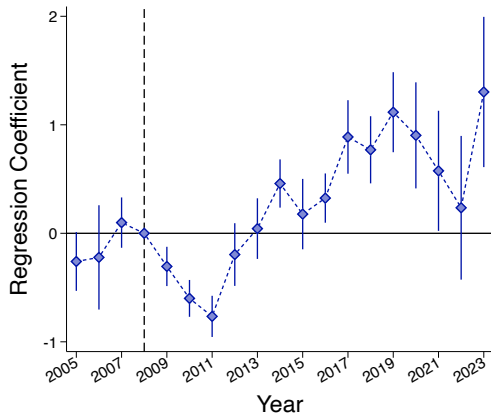
(a) Relative Issuance



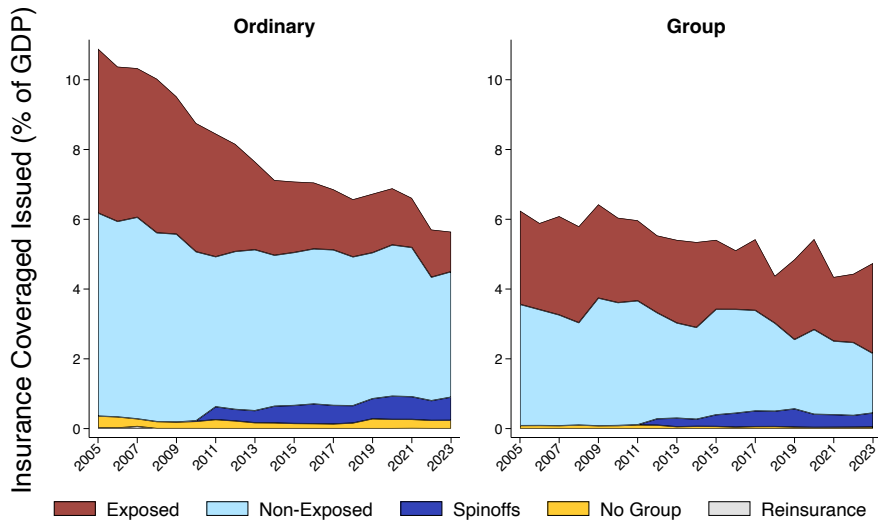
(b) Issuance Shares

Exposed Insurers Transition to Group Policies – Poisson Regression

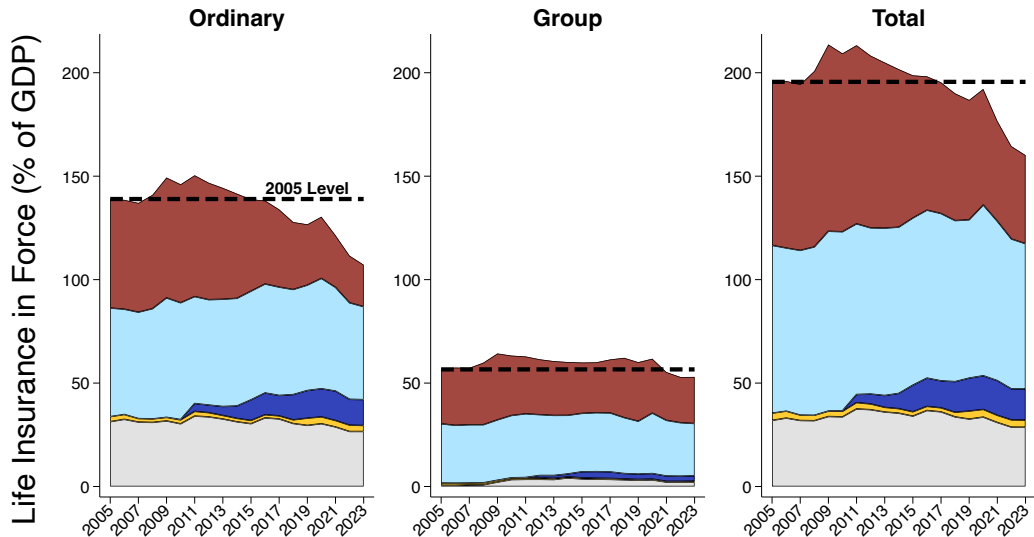
$$\log \mathbb{E}[\text{Issuance}_{ijt}] = \sum_{\tau=2005}^{2023} \beta_{\tau} \mathbf{1}\{\tau = t\} \times \text{Exposed}_j \times \text{Group}_i + \delta_{ij} + \delta_{jt} + \delta_{it} + \varepsilon_{ijt}$$



Exposed Insurers were Responsible for the Aggregate Decline in Issuance



As a Result, The Life Insurance Market Has Shrunk



Conclusion

Interest Rate Risk Matters for Product Markets

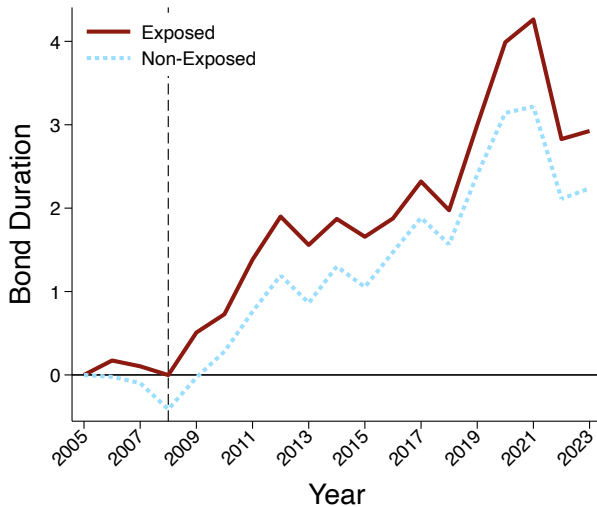
- **Today** — Large swings in product issuance and distortions due to interest rate risk
 - Risk management by financial institutions has major impacts on product markets
 - Large consequences for products with different maturities!
- **Future Work 1** — structural estimation + Counterfactuals
 1. Decomposition of market size into demand + supply characteristics
 2. How would market look today if duration gaps never opened up?
- **Future Work 2** — within-VA-issuer analysis
 1. Certain VAs carry more regulatory IRR exposure than others
 2. Compare exposure by *product* types

Appendix

Summary Statistics for Statutory Filings Data

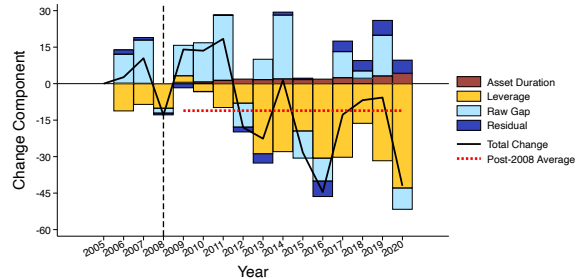
	Exposed Insurers		Non-Exposed Insurers	
	2005-2008	2009-2023	2005-2008	2009-2023
Number of Groups				
Full Sample	26	25	239	198
Compulife Sample	12	15	39	43
Assets	94.68	100.30	8.31	14.57
Surplus	5.09	5.39	0.67	1.25
Leverage Ratio	19.62	19.17	6.56	8.97
Leverage Ratio (Weighted)	20.13	21.15	17.94	16.26
VA Liability Share	0.57	0.50	0.01	0.01
IS Reserve Share	0.67	0.65	0.24	0.25
Issuance Market Share				
Ordinary	0.43	0.29	0.54	0.61
Group	0.45	0.42	0.54	0.51
In Force Market Share				
Ordinary	0.38	0.29	0.37	0.39
Group	0.48	0.44	0.49	0.47

Exposed Insurers Reach for Duration More After the GFC

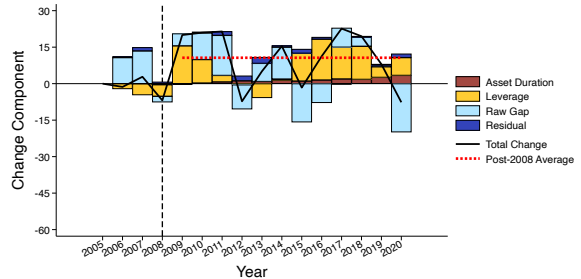


Duration Gap Decomposition

$$\Delta D_{jt}^K = \Delta D_{jt}^A + \Delta \left[\text{Lev}_{jt} \times G_{jt} \right] = \underbrace{\Delta D_{jt}^A}_{\text{Asset Duration}} + \underbrace{\Delta \text{Lev}_{jt} \times G_{jt}}_{\text{Leverage}} + \underbrace{\text{Lev}_{jt} \times \Delta G_{jt}}_{\text{Duration Mismatch}} + \underbrace{\Delta \text{Lev}_{jt} \times \Delta G_{jt}}_{\text{Residual}}$$



(a) Exposed Insurers



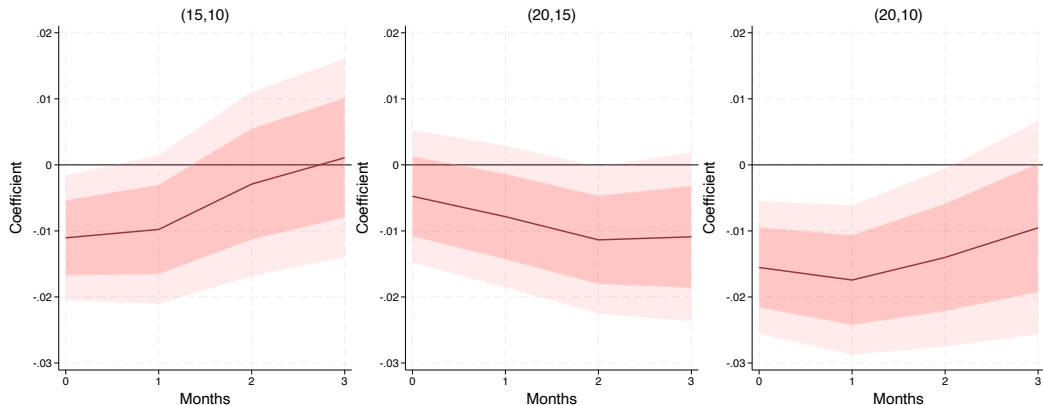
(b) Non-Exposed Insurers

Long-Short Spreads Move more for Exposed Insurers

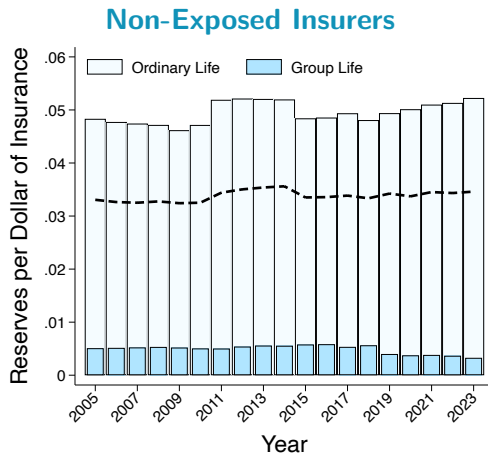
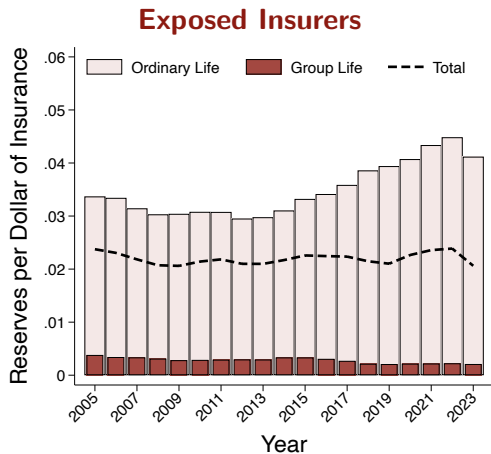
<i>(Long, Short) Category:</i>	(15,10)	(20,15)	(20,10)	(15,10)	(20,15)	(20,10)
$y_t^{(10)} \times \text{Exposed}_j \times \text{Long}_i$	-0.004* (0.002)	-0.015*** (0.002)	-0.019*** (0.003)			
$y_t^{(10)} \times \text{Assets}_{jt} \times \text{Long}_i$	-0.005*** (0.001)	-0.006*** (0.001)	-0.011*** (0.002)			
$\text{MPU}_t \times \text{Exposed}_j \times \text{Long}_i$				0.008*** (0.002)	0.012*** (0.002)	0.020*** (0.003)
$\text{MPU}_t \times \text{Assets}_{jt} \times \text{Long}_i$				0.004*** (0.001)	-0.003*** (0.001)	0.001 (0.001)
Insurer $\times$ Month FE	✓	✓	✓	✓	✓	✓
Insurer $\times$ Product FE	✓	✓	✓	✓	✓	✓
Month $\times$ Product FE	✓	✓	✓	✓	✓	✓
Observations	8956	8956	8956	8956	8956	8956
Within- R^2	0.009	0.044	0.052	0.011	0.023	0.017

Long-Short Spreads Move more for Exposed Insurers

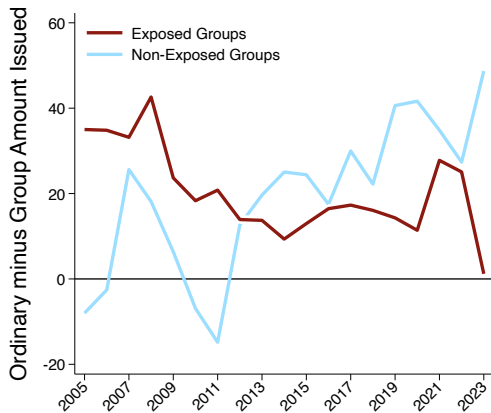
$$\Delta_{(t-3,t+k)} \log \text{Price}_{ij} = \beta \times \Delta_{(t-3,t)} y^{(10)} \times \text{Exposed}_j \times \text{Long}_i + \delta_{jt} + \delta_{it} + \delta_{ij} + \varepsilon_{ijt}$$



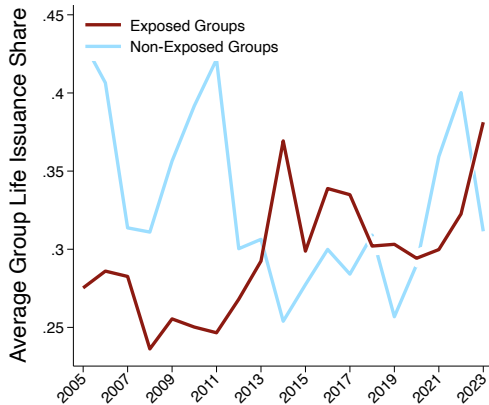
Ordinary life reserves are larger and more interest sensitive than Group



Results Without MetLife



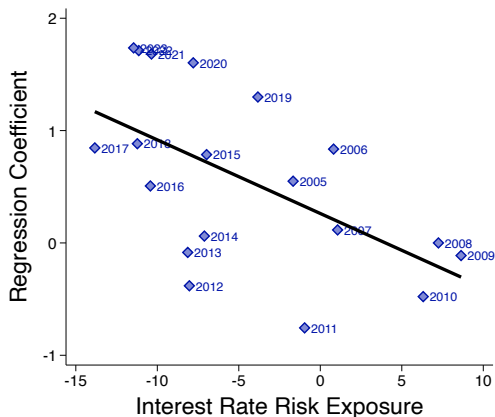
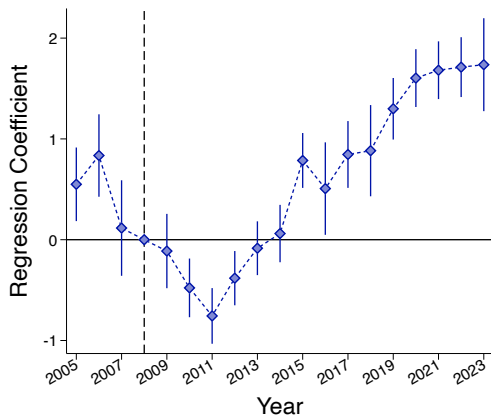
(a) Ordinary Life



(b) Group Life

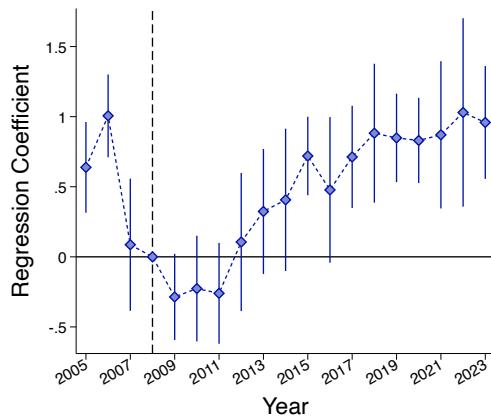
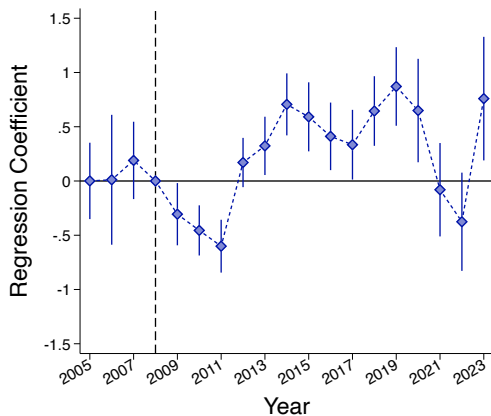
Poisson Regression: Number of Policies

$$\log \mathbb{E}[\# \text{Policies}_{ijt}] = \sum_{\tau=2005}^{2023} \beta_{\tau} \mathbf{1}\{\tau = t\} \times \text{Exposed}_j \times \text{Group}_i + \delta_{ij} + \delta_{jt} + \delta_{it} + \varepsilon_{ijt}$$

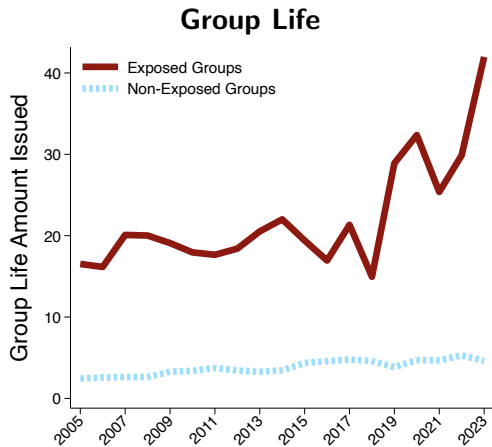
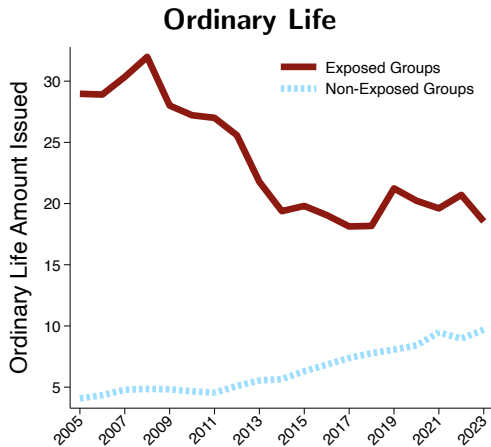


Poisson Regression: Excluding MetLife

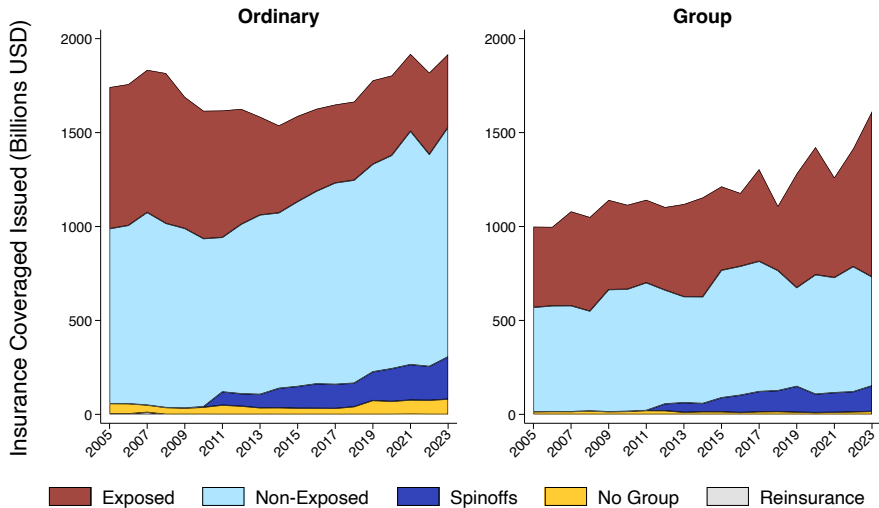
$$\log \mathbb{E}[\#Policies_{ijt}] = \sum_{\tau=2005}^{2023} \beta_{\tau} \mathbf{1}\{\tau = t\} \times \text{Exposed}_j \times \text{Group}_i + \delta_{ij} + \delta_{jt} + \delta_{it} + \varepsilon_{ijt}$$



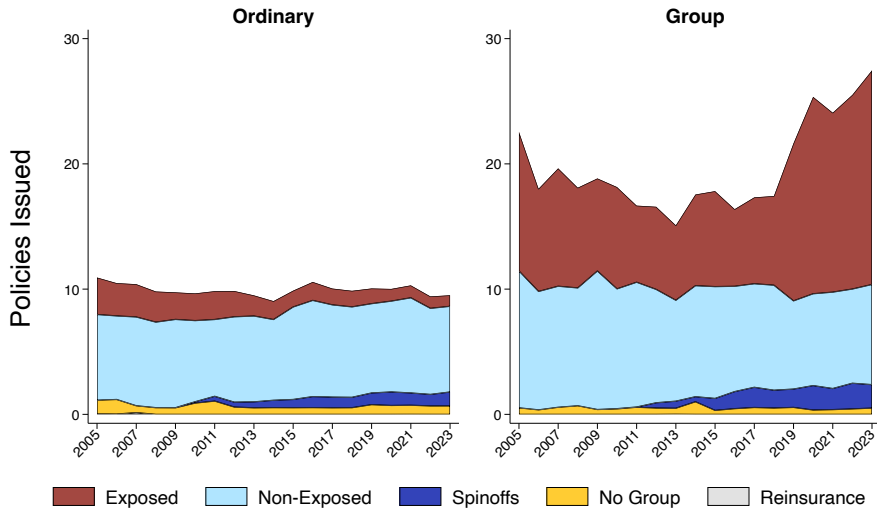
Average Ordinary Issuance Declined (Group Increased) For Exposed Groups



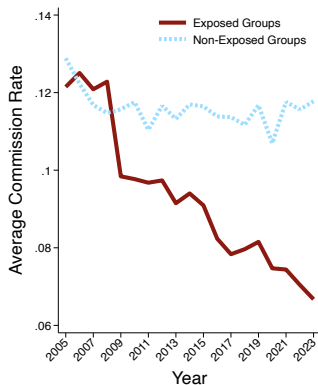
Nominal Ordinary Issuance Steady While Group Life Increases



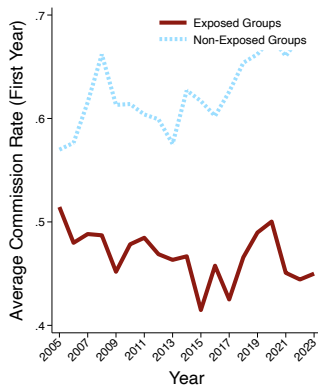
Policy Issuance Followed the Same Trends as Amounts



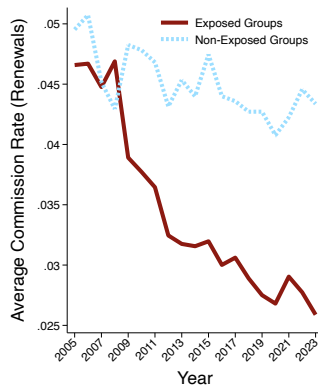
Commissions Followed the Same Trends as Issuance



(a) Total



(b) New Policies



(c) Renewals